

Transformers in AI

Imagine you're reading a book. As you come across a sentence like *"The dog ran after the ball because it was bouncing,"* you automatically know that "it" refers to the "ball," not the dog. This kind of understanding—grasping context, connections, and meanings—is something humans do effortlessly. But for machines, it's incredibly difficult.

That's where **Transformers** come in.

Transformers are a type of deep learning architecture that helps machines understand language (and even images or code) the way we do—by capturing relationships and context between elements in data. Developed by Vaswani et al. in 2017 in the paper *"Attention Is All You Need,"* transformers have since become the foundation of most modern AI systems like GPT, BERT, DALL·E, and more.

The Problem Transformers Solve

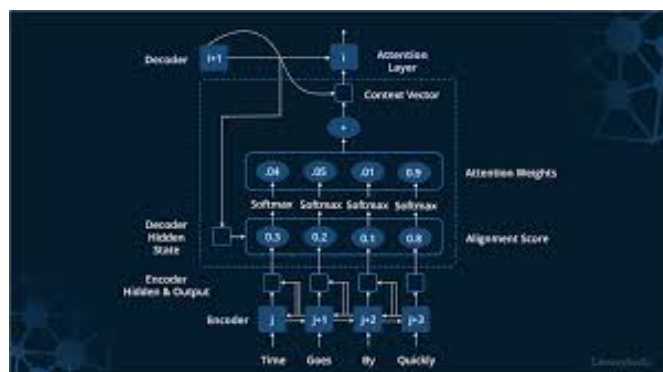
Before transformers, models like **Recurrent Neural Networks (RNNs)** or **Long Short-Term Memory (LSTM)** networks were used to handle sequential data like text. These models processed information one word at a time—like reading one word per second and forgetting parts of the sentence by the time you reach the end.

RNNs are like trying to remember a sentence word by word while only seeing one word at a time, with no ability to glance back. This approach is: slow (can't be parallelized easily), memory-limited (forgets long-range dependencies)

Transformers change the game by **looking at the entire sentence at once** and learning how each word relates to every other word.

Core Concept: Attention (and Self-Attention)

The beating heart of a transformer is a mechanism called **attention**, specifically **self-attention**.



Imagine a party conversation:

You're in a room full of people talking. Even with all the noise, when someone says your name, you instinctively pay attention. You "attend" to the most relevant information—even if it's not the most recent.

This is what self-attention does in transformers. It lets the model decide which words (or parts of the input) to pay more attention to when processing a particular word.

For example, in the sentence:

“The trophy doesn’t fit in the suitcase because it is too small.”

Self-attention helps the model determine that “it” refers to “suitcase”, not “trophy.” This is context-aware understanding.

How Transformers Work: Breaking It Down

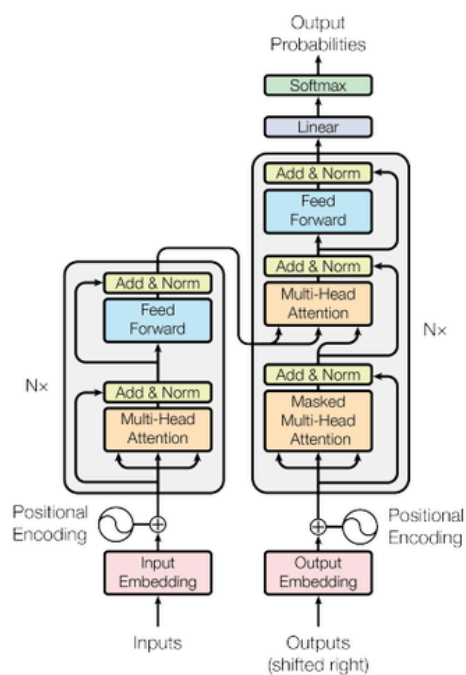


Figure 1: The Transformer - model architecture.

Let’s look at the key building blocks of a transformer:

1. Input Embedding

Each word is turned into a vector (a list of numbers) representing its meaning in a high-dimensional space. These embeddings also include **positional information**, since transformers don’t read data sequentially.

2. Self-Attention Mechanism

Each word compares itself to every other word in the sentence using *query*, *key*, and *value* vectors:

- **Query:** What is this word looking for?
- **Key:** What does each word in the sentence offer?
- **Value:** The actual content of each word.

The model calculates how much attention each word should pay to others.

Think of it like a job interview:

- **Query** is the role you're hiring for.
- **Keys** are applicants' skills.
- **Values** are applicants' resumes.
You rank how well each applicant (word) fits the job and give more weight to the best candidates.

3. Multi-Head Attention

Instead of doing attention once, transformers use **multiple attention heads** that focus on different parts of the sentence. One head might look for grammar, another for meaning, another for subject-object relationships.

Like having multiple experts looking at the same document from different angles.

4. Feedforward Layer

After attention, each word is passed through a neural network layer that processes the information further.

5. Residual Connections + Layer Normalization

These help prevent the model from forgetting earlier information and make training more stable and efficient.

Encoder and Decoder Structure

A standard transformer has two parts:

Encoder

- Reads and understands the input (like a sentence).
- Each encoder layer processes the data through self-attention and feedforward layers.

Decoder

- Generates the output (like translating into another language or predicting the next word).
- Uses attention both on its own output and on the encoder's output.

The encoder is like reading and understanding a recipe; the decoder is like using that understanding to cook the meal.

In models like GPT, only the decoder is used. In models like BERT, only the encoder is used.

Why Transformers Are So Powerful

- **Parallel Processing:** Unlike RNNs, transformers can process entire sequences at once.
- **Context Awareness:** They can relate every word to every other word in the sentence.
- **Scalability:** Works well with massive datasets and deep layers.
- **Transferability:** Once trained on general data, transformers can be fine-tuned for specific tasks (translation, summarization, coding).

Real-World Impact

Today, transformers power:

- **GPT (OpenAI)** – Text generation
- **BERT (Google)** – Search and classification
- **DALL·E** – Text-to-image generation
- **Codex** – Code generation
- **Whisper** – Speech recognition

Their influence extends to fields like healthcare, finance, education, creative arts, and robotics.

Transformers are not just a technical curiosity—they're reshaping how we interact with machines. From chatbots that write essays to systems that can paint pictures or generate code, transformers are making AI more human-like than ever before.

They've taken a fundamental idea—*paying attention*—and turned it into one of the most powerful tools in computing history.